

1. Which notation would you use to denote the 4th layer's activations when the input is the 7th example from the 3rd mini-batch?

1 point

- ☐ $a^{[7]}(3)(4)$
- ☐ $a^{[8]}(7)(4)$
- ☒ $a^{[4]}(3)(7)$

2. Suppose you don't face any memory-related problems. Which of the following make more use of vectorization.

1 point

- ☐ Stochastic Gradient Descent, Batch Gradient Descent, and Mini-Batch Gradient Descent all make equal use of vectorization.
- ☐ Stochastic Gradient Descent
- ☒ Batch Gradient Descent
- ☐ Mini-Batch Gradient Descent with mini-batch size $m/2$.

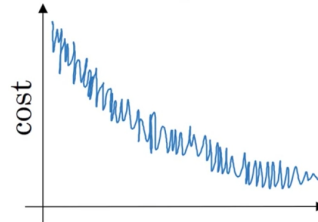
3. We usually choose a mini-batch size greater than 1 and less than n , because that way we make use of vectorization but not fall into the slower case of batch gradient descent.

1 point

- ☒ True
- ☐ False

4. While using mini-batch gradient descent with a batch size larger than 1 but less than m , the plot of the cost function J looks like this:

1 point



You notice that the value of J is not always decreasing. Which of the following is the most likely reason for that?

- ☒ In mini-batch gradient descent we calculate $J(\hat{y}^{(t)}, y^{(t)})$ thus with each batch we compute over a new set of data.
- ☐ A bad implementation of the backpropagation process, we should use gradient check to debug our implementation.
- ☐ The algorithm is on a local minimum thus the noisy behavior.
- ☐ You are not implementing the moving averages correctly. Using moving averages will smooth the graph.

5. Suppose the temperature in Casablanca over the first two days of March are the following:

1 point

March 1st: $\theta_1 = 10^\circ \text{ C}$

March 2nd: $\theta_2 = 25^\circ \text{ C}$

Say you use an exponentially weighted average with $\beta = 0.5$ to track the temperature: $v_0 = 0$, $v_t = \beta v_{t-1} + (1 - \beta) \theta_t$, if v_2 is the value computed after day 2 without bias correction, and $v_2^{\text{corrected}}$ is the value you compute with bias correction. What are these values?

- ☐ $v_2 = 20$, $v_2^{\text{corrected}} = 15$.
- ☒ $v_2 = 15$, $v_2^{\text{corrected}} = 20$.
- ☐ $v_2 = 15$, $v_2^{\text{corrected}} = 15$.
- ☐ $v_2 = 20$, $v_2^{\text{corrected}} = 20$.

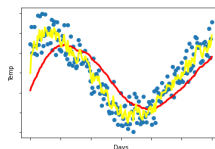
6. Which of the following is true about learning rate decay?

1 point

- ☐ The intuition behind it is that for later epochs our parameters are closer to a minimum thus it is more convenient to take larger steps to accelerate the convergence.
- ☐ It helps to reduce the variance of a model.
- ☐ We use it to increase the size of the steps taken in each mini-batch iteration.
- ☒ The intuition behind it is that for later epochs our parameters are closer to a minimum thus it is more convenient to take smaller steps to prevent large oscillations.

7. You use an exponentially weighted average on the London temperature dataset. You use the following to track the temperature: $v_t = \beta v_{t-1} + (1 - \beta) \theta_t$. The yellow and red lines were computed using values β_1 and β_2 respectively. Which of the following are true?

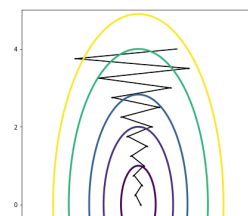
1 point

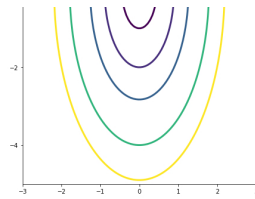


- ☒ $\beta_1 < \beta_2$.
- ☐ $\beta_1 > \beta_2$.
- ☐ $\beta_1 = 0, \beta_2 > 0$.
- ☐ $\beta_1 = \beta_2$.

8. Consider the figure:

1 point





Suppose this plot was generated with gradient descent with momentum $\beta = 0.01$. What happens if we increase the value of β to 0.1?

- ☐ The gradient descent process moves more in the horizontal and the vertical axis.
- ☐ The gradient descent process starts moving more in the horizontal direction and less in the vertical.
- ☒ The gradient descent process moves less in the horizontal direction and more in the vertical direction.
- ☐ The gradient descent process starts oscillating in the vertical direction.

9. Suppose batch gradient descent in a deep network is taking excessively long to find a value of the parameters that achieves a small value for the cost function $\mathcal{J}(W^{[1]}, b^{[1]}, \dots, W^{[L]}, b^{[L]})$. Which of the following techniques could help find parameter values that attain a small value for $\mathcal{J}$? (Check all that apply)

1 point

- ☒ Normalize the input data.
- ☐ Add more data to the training set.
- ☐ Try better random initialization for the weights
- ☒ Try using gradient descent with momentum.

10. Which of the following statements about Adam is *False*?

1 point

- ☐ We usually use "default" values for the hyperparameters β_1 , β_2 and ϵ in Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-9}$)
- ☐ The learning rate hyperparameter α in Adam usually needs to be tuned.
- ☒ Adam should be used with batch gradient computations, not with mini-batches.
- ☐ Adam combines the advantages of RMSProp and momentum